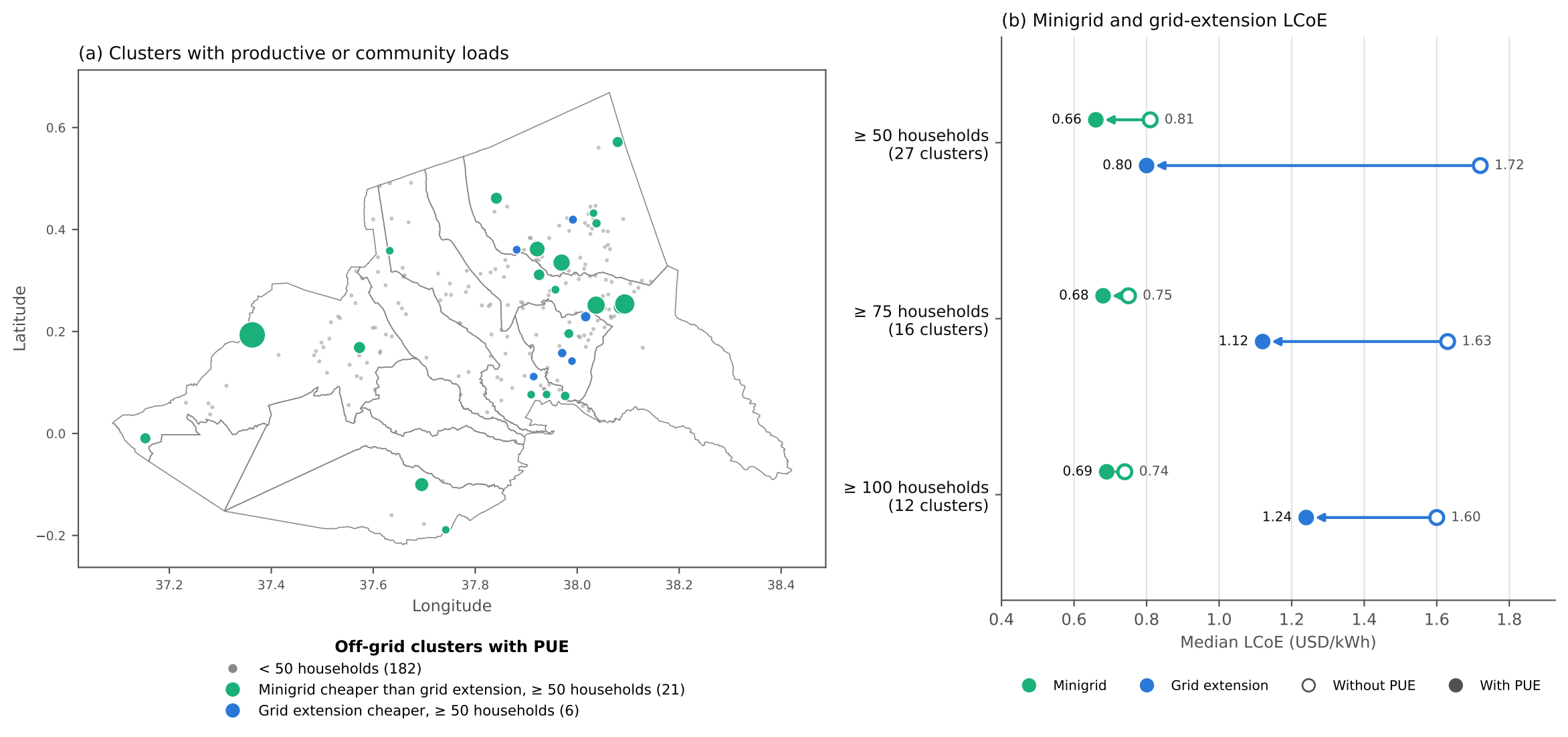

(a) Clusters with productive or community loads
Latitude
Longitude
0.6
0.4
0.2
0.0
−0.2
37.2
37.4
37.6
37.8
38.0
38.2
38.4
Off-grid clusters with PUE
< 50 households (182)
Minigrid cheaper than grid extension, ≥ 50 households (21)
Grid extension cheaper, ≥ 50 households (6)
(b) Minigrid and grid-extension LCoE
≥ 50 households
(27 clusters)
0.66
0.81
0.80
1.72
≥ 75 households
(16 clusters)
0.68
0.75
1.12
1.63
≥ 100 households
(12 clusters)
0.69
0.74
1.24
1.60
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Median LCoE (USD/kWh)
Minigrid
Grid extension
Without PUE
With PUE